# Ogbonna Arinze

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## SUMMARY

Senior Full-Stack Engineer with 8 years of professional experience designing, building, and maintaining business-critical systems across healthcare, fintech, civic tech, and enterprise environments. Deep expertise in **TypeScript**, **React**, **Next.js**, **NestJS**, and **Node.js**, with a proven track record of owning features end-to-end from UX to deployment. Experienced in API design, data modeling, system integration (REST/GraphQL), and championing clean architecture, reusable UI patterns, and engineering best practices. Pragmatic, product-focused engineer who thrives in autonomous, outcome-oriented teams.

## CORE COMPETENCIES

- **Full-Stack Development:** TypeScript, React, Next.js (SSR/SSG/ISR), NestJS, Node.js, and end-to-end feature ownership from UI to API to deployment.
- **API Design & Data Modeling:** RESTful APIs, GraphQL, PostgreSQL, MongoDB, Prisma ORM, JWT authentication, RBAC, and system integration patterns.
- **Frontend Architecture:** Scalable component systems, design systems (Radix UI, Tailwind CSS), monorepos (Nx, pnpm workspaces), and reusable UI patterns.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, and CI-driven QA pipelines.
- **Cloud & Infrastructure:** Google Cloud, AWS (CloudFront, S3), Docker, GitHub Actions, CI/CD pipelines, and Infrastructure as Code.
- **Performance & Observability:** Core Web Vitals, code splitting, CDN strategies, Lighthouse, Sentry, Datadog RUM, and bundle analysis.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Mastra, vector search (pgvector, Chroma), and agentic workflows.
- **Leadership:** Technical direction, architecture reviews, mentorship, Agile/Scrum, TDD/BDD, and product-minded development.

## WORK EXPERIENCE

**Genentech (Roche)** — San Francisco, CA (Remote)

***Frontend Engineer*** — *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, optimizing workflows for clinical site integrations with upstream EHR systems and enhancing adoption.
- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query their data in real time via a conversational interface.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Implemented **JWT-based authentication** and **role-based access control** in **Next.js** with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Optimized application performance by implementing **lazy loading, code splitting**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

**Descasio** — Lagos, Nigeria (Hybrid)

***Senior Frontend Developer*** — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with software architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application, significantly boosting speed and user experience.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports and improving code reliability.
- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices with precise user story planning.

- Established engineering standards for code quality, testing, and deployment using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers and guided frontend best practices, fostering high-quality, maintainable code across projects.

**ActivEdge Technologies** Lagos, Nigeria
***Senior React Developer*** *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3 frontend engineers, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React** with **Redux** for state management, **RxJS** for data streaming, and **Axios** for asynchronous API requests.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design** for user role and permission management.

**Founders Lair** Beijing, China (Remote)
***Full Stack Engineer (Contract)*** *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, ensuring technical solutions aligned with business needs.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** (testing) and **AWS** (production).
- Mentored two interns through frontend and backend projects, providing technical guidance and code review across the full stack.

**SureTrade Business Solutions** Nigeria
***Frontend Developer*** *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

## PROJECTS

**OurNigeria** ournigeria.ng
***Senior Engineer*** *Feb 2026 - Present*

*AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.*

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app (22 interactive chart types with Recharts), **Next.js** admin dashboard, **NestJS** document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with **pnpm workspaces**.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including pgvector extension for vector storage, and built a multi-agent AI system using **Mastra** with 5 specialized agents and a hybrid RAG pipeline combining **pgvector** + BM25 full-text search with Reciprocal Rank Fusion and Cohere reranking.
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats, token-budgeted context management, phone OTP and Telegram OAuth authentication, and shareable public conversations with SEO optimization (Open Graph, JSON-LD, SSR).
- Built the document ingestion pipeline supporting PDF (with OCR via Tesseract), XLSX/XLS, DOCX, JSON, and Markdown extractors, processing 700+ documents (959 files) across 37 states with zero ingestion errors.
- Configured CI/CD with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and **Traefik** for reverse proxy routing.

## EDUCATION

**University of Nigeria** Nsukka, Enugu
*BSc Computer Science - 4.1 CGPA* *Graduation Date: Sep 2019*